Causal inference on endogenous social network formation

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Motivation

- Social networks shape key economic and social outcomes:
  - Diffusion of (mis)information,
  - access to jobs and opportunities,
  - spread of financial shocks and disease.
- Large empirical literature: network features (density, centrality, size) are highly predictive of individual behavior and social outcomes.
- We know less about the causal mechanisms driving the *formation* of social ties.
- Theory suggests various endogenous mechanisms:
  - *Triadic closure*: friends-of-friends become friends.
  - *Cumulative advantage*: high-degree nodes attract more ties.
  - *Local density*: dense neighborhoods may inhibit new ties.

Methodological challenges

- Why are there so many triangles?
    1. Homophily based on unobservables? (confounding)
    2. Causal effect of indirect ties?
       $\Rightarrow$ Which tie is the chicken, which the egg? (reflection problem)
- When is a network in equilibrium?
    1. No desired individual deviations? (Nash)
    2. No pairwise coordinated deviations? (pairwise stability)
    - Pervasive equilibrium multiplicity!
- What is the population / estimand of interest?
    1. Hypothetical infinite super-network?
    2. Population of disconnected networks?

This paper

- *Our approach*: design-based framework requiring
  - *Random initial assignment*: initial ties permuted across some individuals.
  - *Panel data*: network observed for (at least) two time periods.
- *Estimands*:
  - Sample average treatment effects (SATEs)
    of initial network structure on tie formation.
  - No equilibrium model, no superpopulation needed.
- *Application*: professional services firm;
  new hires randomly assigned to project teams within offices.
- *Finding*: indirect ties have a strong (+40%), robust effect;
  degree and local density effects are smaller and less robust.

Literature

1. **Mechanisms of tie formation**
   - *Triadic closure*: Simmel and Wolff (1964), Coleman (1988), Heider (2013).
   - *Cumulative advantage*: Merton (1968), Barabasi and Albert (1999), DiPrete and Eirich (2006), DiMaggio and Garip (2012).
   - *Local density*: Burt (2009), Uzzi (1997), Gargiulo and Benassi (2000).
2. **Statistical network models**: (Kolaczyk and Csárdi, 2014)
   - Exponential random graph models.
   - Stochastic block models.
   - Dynamic models.
3. **Structural econometric network models**: Graham (2015), De Paula et al. (2018), Graham and De Paula (2020), Christakis et al. (2020).
4. **Design-based inference**:
   Neyman (1923), Abadie et al. (2020), Abadie et al. (2023).

Setup

Identification and inference

Empirical application

Discussion

Setup and notation

- Time periods $t = 1, 2$, individuals $i, j \in \{1 \dots n\}$.
- Adjacency matrices A^t with $A^t_{ij} \in \{0, 1\}$.
- Structural (causal) relationship $A^2 = f(A^1)$.
- Randomization of initial network: A^1 uniform from $\mathcal{A}$.
- Design-based identification and inference:
  - We condition on sample $\{1 \dots n\}$, and on potential outcomes f.
  - Only source of randomness: Sampling of A^1 from $\mathcal{A}$.

Assumption 1 (Setup)

- *Structural relationship*: $A^2 = f(A^1)$.
- *Repeated observation*: Both A^1 and A^2 are observed.
- *Randomization*: $P(A^1 = A|f) = \frac{1}{|\mathcal{A}|}$ for all $A \in \mathcal{A}$.
- *Exclusion restriction*: $d_{ij}(A) = d \Rightarrow y_{ij}(f(A)) = Y_{ij}^d$ for all $A \in \mathcal{A}$.
- *Support*: $P(d_{ij}(A^1) = d) > 0$.

Examples

- Outcome: $Y_{ij} = A^2_{ij}$ - presence of a tie between i and j.
- Treatments: $D_{ij} = d_{ij}(A^1)$.
  - *Indirect tie*:
$$D_{ij} = \mathbf{1}\left(\sum_k A^1_{ik} A^1_{kj} > 0\right)$$
  - *Network degree*:
$$D_{ij} = \sum_k A^1_{ik}$$
  - *Local network density*:
$$D_{ij} = \frac{\sum_{k>k'} A^1_{ik} A^1_{kk'} A^1_{k'i}}{\sum_{k>k'} A^1_{ik} A^1_{k'i}}$$
- Randomization:
  $\mathcal{A}$ is the set of matrices obtained by swapping new hires within offices.

Assumption 2 (Randomization by permutation)

1. *Permutations*: For any permutation π of $\{1, \ldots, n\}$, denote A_π the matrix with (i, j)th entry $A_{\pi(i),\pi(j)}$. Let Π be an algebraic group of permutations. The set $\mathcal{A}$ is given by
$$\mathcal{A} = \{A_\pi : \ \pi \in \Pi\}.$$

2. *Invariance and equivariance*: For all $\pi \in \Pi$, $\mathcal{E}$ is invariant under π,
$$(i, j) \in \mathcal{E} \Rightarrow (\pi(i), \pi(j)) \in \mathcal{E},$$
and for all $\pi \in \Pi$ and $(i, j) \in \mathcal{E}$, d_{ij} is equivariant under π :
$$d_{ij}(A_\pi) = d_{\pi(i),\pi(j)}(A).$$

Assumption 3 (Permutation of subset of nodes)

- Denote $\mathcal{S}_n$ is the *symmetric group* of all permutations of the set $1, \ldots, n$.
- The permutation π is drawn from

$$\begin{aligned}\Pi = \{\pi \in \mathcal{S}_n : \pi(j) = j \quad \forall j \notin \mathcal{I}, \\ O(\pi(i)) = O(i) \quad \forall i \in \mathcal{I}\}.\end{aligned}$$

- In the application, only *new hires* are randomly assigned; senior employees stay fixed.

Lemma 1: Inverse probability weighting

- Denote $Y_{ij} = y_{ij}(A^2)$ and $D_{ij} = d_{ij}(A^1)$.
- Under Assumption 1, the IPW estimator

$$\widehat{Y}_{ij}^{d} = Y_{ij} \cdot \frac{\mathbf{1}(D_{ij} = d)}{P(D_{ij} = d)}$$

satisfies

$$E\left[\widehat{Y}_{ij}^{d} | f\right] = Y_{ij}^{d}.$$

Lemma 2: Equivariance of randomization probabilities

- Suppose that item 3 of Assumption 1 (randomization) and Assumption 2 hold.
- Denote $p_{ij}(d) = P(D_{ij} = d)$, and $P_{ij} = p_{ij}(D_{ij})$.
- Then for all $\pi \in \Pi$

$$p_{\pi(i)\pi(j)}(d) = p_{ij}(d), \quad \text{and} \quad P_{\pi(i)\pi(j)}|f \sim P_{ij}|f.$$

Proposition 1: Inverse probability weighted linear regression

- Denote $p_{ij}(d) = P(D_{ij} = d)$, and $P_{ij} = p_{ij}(D_{ij})$.
- Define

$$\widehat{\beta} = \underset{b}{\operatorname{argmin}} \sum_{(i,j)\in\mathcal{E}} \frac{1}{P_{ij}} \left(Y_{ij} - D_{ij} \cdot b\right)^2,$$

$$\beta = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \left[\left(\sum_{d\in\mathcal{D}} d \cdot d' \right)^{-1} \cdot \left(\sum_{d\in\mathcal{D}} Y_{ij}^d \cdot d \right) \right].$$

- Under Assumptions 1 and 2,

$$E[\widehat{\beta}|f] = \beta.$$

Sample average treatment effect

- Special case: Binary treatment.
- $D_{ij} = (1, X_{ij})$, with $X_{ij} \in \{0, 1\}$.
- Then

$$\beta_2 = \sum_{(i,j)\in\mathcal{E}} (Y_{ij}^1 - Y_{ij}^0)$$

  is the sample average treatment effect.
- Support condition: For all $(i, j) \in \mathcal{E}$,

$$0 < P(X_{ij} = 1) < 1.$$

- Example: Triadic closure.

Proposition 2: Within regression

- Suppose $\mathcal{E} = \mathcal{I} \times \mathcal{J}$, where $\mathcal{I} \cap \mathcal{J} = \emptyset$.
- Consider the multisets $\mathcal{D}_j = [D_{ij} : \ i \in \mathcal{I}]$.
- Define:

$$\widehat{\gamma}_j = \operatorname*{argmin}_c \sum_{i \in \mathcal{I}} (Y_{ij} - D_{ij} \cdot c)^2 , \qquad \widehat{\gamma} = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \widehat{\gamma}_j,$$

$$\gamma_j = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \left[\left(\sum_{d \in \mathcal{D}_j} d \cdot d' \right)^{-1} \cdot \left(\sum_{d \in \mathcal{D}_j} Y_{ij}^d \cdot d \right) \right], \quad \gamma = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \gamma_j .$$

- Under Assumptions 1, 2, and 3:

$$E[\widehat{\gamma} | f] = \gamma.$$

Computational implementation (IPW point estimate)

- Under Assumption 3, store data in matrices of dimension $|\mathcal{I}| \times |\mathcal{J}|$:

$$Y = (A_{ij}^2)_{i\in\mathcal{I},j\in\mathcal{J}}, \quad D = (D_{ij})_{i\in\mathcal{I},j\in\mathcal{J}}.$$

- Assignment probabilities:

$$p_{ij}(d) = \tfrac{1}{|\mathcal{I}|}\sum_{i'} \mathbf{1}(D_{i'j} = d), \qquad P_{ij} = p_{ij}(D_{ij}). \tag{1}$$

- "Instrument:" $Z_{ij} = \frac{1}{P_{ij}} D_{ij}$.
- Weighted regression:

$$C = \left(\sum_{i\in\mathcal{I},j\in\mathcal{J}} Z_{ij} \cdot D'_{ij}\right)^{-1}, \quad B_{i,i'} = C \cdot \left(\sum_{j\in\mathcal{J}} Z_{ij} \cdot Y_{i'j}\right), \quad \widehat{\beta} = \sum_{i\in\mathcal{I}} B_{i,i}. \tag{2}$$

Proposition 3: Randomization inference

- Null hypothesis:
  Y_{ij}^d does not depend on d for all $(i, j) \in \mathcal{E}$ (sharp null of no effects).

- *IPW version* (Prop. 1 setting): permuted estimator and p-value

$$\widehat{\beta}_\pi = \underset{\beta}{\operatorname{argmin}} \sum_{(i,j)\in\mathcal{E}} \frac{1}{P_{\pi(i)\pi(j)}} \left(Y_{ij} - D_{\pi(i)\pi(j)} \cdot \beta\right)^2,$$

$$p^\beta = \frac{1}{|\Pi|} \sum_{\pi\in\Pi} \mathbf{1}\left(\widehat{\beta} \leq \widehat{\beta}_\pi\right).$$

- Under the null, $P(p \leq \alpha) \leq \alpha$.

Randomization inference continued

- *Within-regression version* (Prop. 2 setting):

$$\widehat{\gamma}_\pi = \frac{1}{|J|} \sum_{j \in \mathcal{J}} \operatorname*{argmin}_c \sum_{i \in \mathcal{I}} \left(Y_{ij} - D_{\pi(i)j} \cdot c\right)^2 ,$$
$$p^\gamma = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}\left(\widehat{\gamma} \leq \widehat{\gamma}_\pi\right) .$$

- No asymptotic approximation needed: Exact finite-sample validity under the permutation distribution.

Computational implementation (randomization inference)

- Permutations π:
  Column j remains constant, any permutation of the rows i is allowed.
- Randomization inference:
  - Do not need to re-calculate the terms C and $B_{i,i'}$.
  - The permuted estimator $\widehat{\beta}_\pi$ is simply given by

$$\widehat{\beta}_\pi = \sum_{i \in \mathcal{I}} B_{\pi(i),i}.$$

- Aggregation across offices o:

$$\widehat{\beta} = \sum_o \frac{m_o}{m} \cdot \widehat{\beta}_o.$$

Approximate normality

- Berry–Esseen bounds for stratified permutation distributions (from Tian et al., 2025).
- $\widehat{\beta}$ and $\widehat{\gamma}$ are approximately normally distributed, conditional on f, over draws of π.
- E.g., there exists a universal constant C such that

$$\sup_{t \in \mathbb{R}} \left| P(\widehat{\beta} \leq \beta + \sigma_\beta \cdot t | f) - \Phi(t) \right| \leq \frac{C}{\sigma_\beta^3} \cdot \sum_o \frac{m_o^2}{m^3} \sum_{i,j \in \mathcal{I}_o} |\tilde{B}_{ij}|^3.$$

- The bound is *negligible for many offices, or for large offices*.
- Similarly, $\widehat{\beta}_\pi$ and $\widehat{\gamma}_\pi$ are approximately normal under the sharp null.

Conservative standard errors and confidence intervals

- σ_β^2 and σ_γ^2 are *not* identified in general.
    - They depend on treatment effect heterogeneity; cf. Neyman, 1923.
    - Upper bounds are identifed (Mukerjee et al., 2018).
- We can estimate *upper bounds* using between-office variance:

$$\widehat{V}_\beta = \tfrac{N}{N-1} \sum_o \left(\tfrac{m_o}{m} \widehat{\beta}_o - \tfrac{\widehat{\beta}}{N} \right)^2, \quad E\left[\widehat{V}_\beta | f \right] \geq \sigma_\beta^2.$$

- The bound is sharp if treatment effects do not vary across offices.
- Together with approximate normality, this yields *conservative confidence intervals*:

$$CI_{1-\alpha} = \left[\widehat{\beta} \pm z_{1-\alpha/2} \cdot \sqrt{\widehat{V}_\beta} \right].$$

Empirical setting

- Global firm in the professional services industry.
- Entry-level employees hired straight from degree programs.
- Work in project teams. Tie $\approx$ working together.
- Initial team assignment determined by an HR manager.
  Random within offices.
- Later team assignment based on an internal labor market.
  Junior employees aim to be recruited by senior colleagues.

Definition of treatment variables

Variable	Definition
CONTINUOUS	
N indirect ties	Number of indirect ties between i and j.
Degree	Network degree of i.
Local density	Local network density. Scaled in percentage points.
BINARY	
Indirect tie	Indicator for whether an indirect tie was present (*N indirect ties* > 0).
High degree	*Degree* greater than the sample median of 75.
High density	*Local density* greater than the sample median of 63.

Sample characteristics

- 6,042 new hires in 979 offices; 130M+ pairs (i, j).

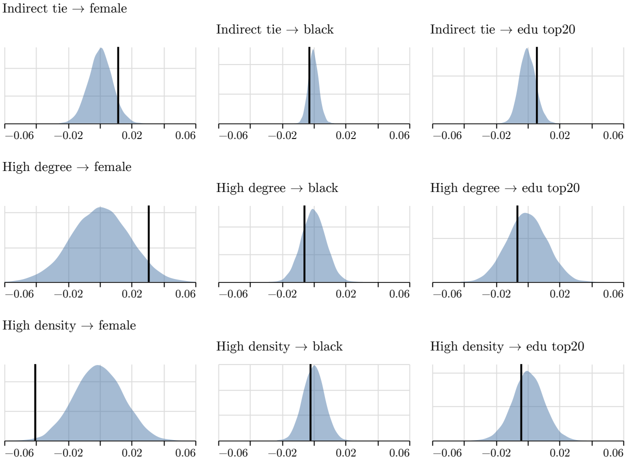
Variable	Mean	Std dev
Tie formed	0.004	0.065
Indirect tie	0.385	0.487
High degree	0.507	0.500
High density	0.501	0.500
N indirect ties	1.573	4.539
Degree	88.120	62.487
Local density	65.627	19.352
Female	0.461	0.499
Black	0.048	0.214
Edu top20	0.113	0.317

Placebo tests

Treatment \ Outcome	Female	Black	Edu top20
Indirect tie	0.012 [0.066]	−0.003 [0.797]	0.006 [0.126]
High degree	0.031 [0.046]	−0.006 [0.774]	−0.006 [0.685]
High density	−0.041 [0.997]	−0.002 [0.618]	−0.004 [0.676]

- P-values in [].
- None significant after Bonferroni correction:
  Cutoffs $.0028$ and $.9972$ for two-sided test at 5% across 9 regressions.

Placebo tests: Permutation distributions



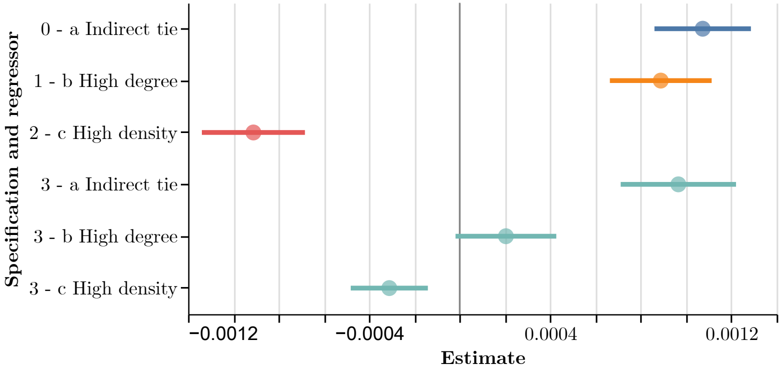
Point estimates fall within the permutation distributions.
Distributions appear normal, confirming the Berry–Esseen bounds.

Effect estimates: Binary treatments

N hires	Intercept	Indirect tie	High degree	High density	Edges
6,042	0.0026	0.00107 (0.00011) [0.000]			130.7M
4,414	0.0038		0.00089 (0.00011) [0.000]		106.0M
5,161	0.0051			-0.00091 (0.00012) [1.000]	118.7M
3,601	0.0023	0.00097 (0.00013) [0.000]	0.00020 (0.00011) [0.029]	-0.00031 (0.00009) [0.999]	85.1M

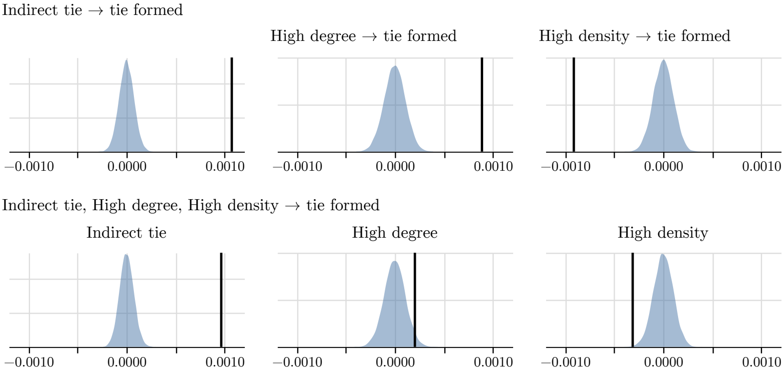
Conservative standard errors in (), one-sided p-values in [].

Confidence intervals: Binary treatments



Based on conservative standard errors and approximate normality.

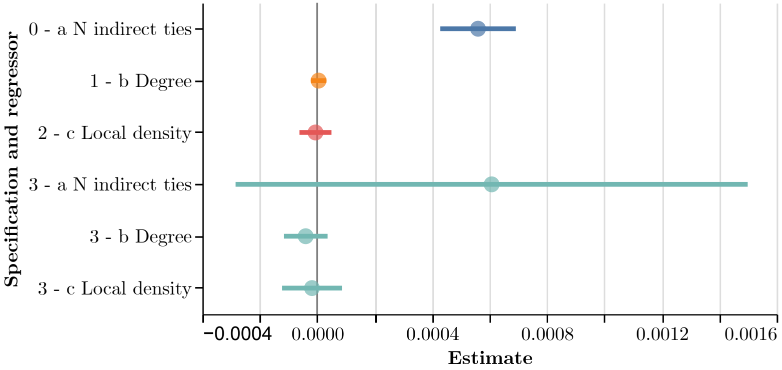
Effect estimates: Permutation distributions (binary)



Effect estimates: Continuous treatments

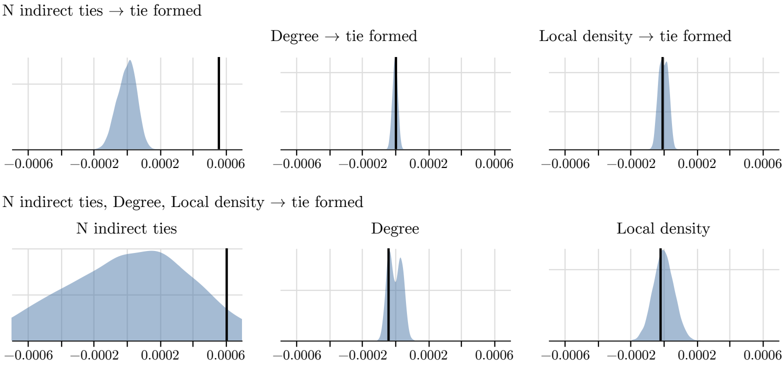
N hires	Intercept	N indirect ties	Degree	Local density	Edges
6,042	0.0040	0.00056 (0.00007) [0.000]			127.7M
6,038	0.0034		0.00000 (0.00001) [0.386]		130.6M
6,036	0.0059			-0.00001 (0.00003) [0.562]	130.7M
4,768	0.0065	0.00061 (0.00045) [0.056]	-0.00004 (0.00004) [0.791]	-0.00002 (0.00005) [0.617]	113.3M

Confidence intervals: Continuous treatments



Based on conservative standard errors and approximate normality.

Effect estimates: Permutation distributions (continuous)



Setup

Identification and inference

Empirical application

Discussion

Discussion (1)

- Indirect ties, network degree, and local density all appear to matter for tie formation.
- The effect of indirect ties is quite robust and highly significant.
- The effects of network degree and local density are less robust.

Discussion (2)

- *Design-based identification and inference*:
  - Avoids awkwardness of sampling models for networks:
    Many small networks? Infinite super-network?
  - Avoids arbitrary asymptotics.
  - Based solely on partial randomness of initial ties.
- *Heterogeneity, super-populations, and estimands*:
  - No need to assume treatment effects are the same across ties.
  - No need to assume hypothetical super-population.
  - Inference for variants of the sample average treatment effect.

Discussion (3)

- *Dynamics versus equilibrium*:
  - Equilibrium notions for networks are ambiguous.
  - Who needs to consent to tie formation? Tie disolution?
  - Additionally: Many equilibria - equilibrium selection?
  - Focusing on network dynamics allows us to avoid taking a stance.
- *Confounders and reverse causality*:
  - Confounders: Unobserved heterogeneity can easily lead to patterns like triadic closure, cumulative advantage.
  - Reverse causality / reflection problem:
    Did the tie between $1, 2$ cause the tie between $2, 3$, or the other way around?
  - Random initial assignment solves both.

Thank you!